Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 17)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

0 1 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ____ %

0 1 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

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0 1 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

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0 1 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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0 2 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 2 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

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0 2 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 3 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 3 . 1 Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 3 . 2 Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

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0 3 . 3 Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

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0 3 . 4 Explain why bird wing bones are a useful model for engineers.

[1 mark]

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0 4 Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha^{-1}	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

0 4 . 1 Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

0 4 . 2 Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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0 4 . 3 Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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0 5 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 5 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 5 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 6 The Isle of Eigg (area 31 km^2, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 6 . 1 Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

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0 6 . 2 Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

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0 6 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

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0 6 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

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0 7 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO_2, NOx

0 7 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 7 . 2 Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 7 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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0 8 Table 1 shows some designated protected areas in the UK.

0 8 . 1 Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

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0 8 . 2 Explain how designating a protected area protects habitats and species.

[3 marks]

0 8 . 3 Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

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0 9 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 9 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 9 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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1 0 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

1 1 . 2 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS